OpenSees Nonlinear Modeling Input Sheet

Models A and B, Bangkok State Audit Office Study

This is Phase 2, the nonlinear analysis in OpenSees (OpenSeesPy). It uses the **same building, the same 27 parameter combinations and the same two framings** as the ETABS Elastic Modeling Input Sheet. Model A is the as-built flat slab on the eccentric core; Model B is the proposed column-beam-slab frame. Geometry, materials, supports and seismic input are identical in both; the gravity framing system is the only variable.

The elastic sheet answers *how large are the demands*. This sheet answers *what yields, what fails, in what order, how the load redistributes, and at what shaking the building collapses*. Section 0 lists what changes; the rest gives the inputs.

0. What changes from the elastic sheet

Item	ETABS elastic (Phase 1)	OpenSees nonlinear (Phase 2)
Geometry, grid, 27-run matrix, supports	As defined	**Unchanged.** Same values, Sections 1-2 below.
Materials	E and Poisson's ratio only	Full stress-strain curves: concrete crushing and cracking (confined / unconfined), rebar yield, buckling and fracture. Section 3.
Reinforcement	Not needed	**Required.** Every wall, column, beam and link beam needs bar areas, because strength comes from them. Section 4.
Stiffness modifiers (0.25, 0.35, 0.15 ...)	Assigned by hand (ACI 6.6.3.1.1)	**Removed** where cracking is modelled by the fibres. Kept only for the elastic middle parts of members. Section 5.
Elements	Shells and frames	Fibre wall piers with shear springs, plastic-hinge columns and beams, link beams with an anchorage (bond) failure, slab strips with punching springs. Section 5.
Seismic input	Response spectrum (RS-DPT61-Z5, RS-2025-REC) with CQC and SRSS	**Ground-motion time histories**: acceleration records applied step by step, X and Y together. The response spectrum is no longer the load. Section 7.
Response spectrum	Is the load case	**Still used, but in the background:** (1) target the records are selected or matched to; (2) design spectrum for sizing the reinforcement; (3) the DBE / MCE levels the records are scaled to.
R factor and I/R x g scale factor	1.635 (R = 6), 1.226 (R = 8), 9.81 (recorded)	**Removed.** Ductility is modelled directly, so the records are applied unreduced. Intensity is set by a scale factor on the record (SF), where SF = 1 is the 2025 event.
Gravity	Mass source only (DEAD); P-Delta from mass	**Gravity is applied first** as a real load (D, plus SDL and 0.25 L if adopted). Wall axial load and slab shear at each column control failure.
Damping	5 % modal, built into the spectrum	**2.5 % Rayleigh** (tall building), explicit in the analysis. Section 8.
Analyses	Modal + response spectrum	Modal (check against ETABS), gravity, pushover, nonlinear time history and incremental dynamic analysis (IDA), sudden core-pier removal, pushdown. Section 9.
Results	Forces, drifts, demand/capacity	Damage sequence, residual drift, load redistribution paths, collapse yes/no, **collapse intensity**. Section 10.
Deficiencies	f'c only (as a matrix level)	f'c levels **plus** link-beam embedment (bar slip), which ETABS cannot model, and the non-compliant design option. Section 6.

1. Fixed Geometry (unchanged from ETABS sheet Section 1)

Parameter	Value	Note for OpenSees
Building height	137.0 m, 33 storeys	One floor node set per storey
Ground floor / typical storey	6.000 m / 4.094 m	Storey elevations as ETABS sheet 1.1
Building grid	39.0 x 39.0 m	
Column grid	7 lines each way at 6.50 m (0 ... 39.0)	As adopted in ETABS sheet 2.3
Core	17.0 x 17.0 m; X 11.0-28.0 m, Y 22.0-39.0 m; on the rear edge	Eccentric: torsion is captured through the rigid-diaphragm master node
Core wall thickness	250 mm	
Coupling (link) beam	250 x 800 mm, clear span 2.4 m (ln/h = 3.0)	Opening width as ETABS sheet Section 9
Beams (Model B only)	500 x 800 mm on every column line	T-beam with slab flange
Columns	1400 sq (F1-10), 1300 sq (F11-20), 1200 sq (F21-28), 800 dia (F29-33)	

2. Iteration Matrix (unchanged) and run count

The same three parameters at three levels each (27 combinations), for Model A and Model B (54 models). Run identifiers as in the ETABS sheet: `1A-ABC` is Model A, solid core, 4,000 psi, 250 mm slab.

Parameter	Level A	Level B	Level C
Core configuration	Solid, no openings	2 openings (front face)	4 openings (front + sides)
Core concrete f'c	3,000 psi (21 MPa)	4,000 psi (28 MPa)	5,000 psi (35 MPa)
Slab thickness	150 mm	200 mm	250 mm

What is new: each model is run many times, not once. An ETABS response-spectrum case is one run per model. A nonlinear time history is one run per record per intensity. Recommended staging:

Stage	Runs	What it answers	Compute (4 cores)
1. All 54 models	Modal, gravity, pushover X and Y, 3 records at SF = 1 (2025 event)	Did any configuration fail in the actual event? Strength and ductility of every combination	about 1.5-2 days
2. All 54 models	3 records scaled to DPT DBE and MCE at T1	Code-level performance (replaces the RS-DPT load cases)	about 3 days
3. Selected models	IDA: SF 0.5 to 8, bisection to 20 %	Collapse intensity: how much stronger than 2025 the shaking must be	about 0.5 day per model
4. Selected models	Sudden core-pier removal and pushdown	Alternate load paths and progressive collapse	minutes per run

Records of about 150 s take 30 to 60 minutes each in a 33-storey 3-D model. Run Stage 1 first; choose Stage 3 and 4 models from its results (for example the extremes and the mid-level 1A/1B-BBB).

3. Materials (nonlinear stress-strain)

Keep the ETABS names and switch the assignment between runs, never edit a material in place. In OpenSees each name becomes a set of uniaxial materials used in the fibre sections.

Name	OpenSees material	Parameters	Use
C-CORE-3000 / 4000 / 5000 unconfined	Concrete02 (Kent-Park, linear tension softening)	f'c = 21 / 28 / 35 MPa; peak at 0.002; 20 % f'c residual at 0.006; ft = 0.17 sqrt(f'c); tension softening to zero at ft/Ets = 0.002	Wall cover and wall web; whole wall if unconfined (non-compliant design option)
C-CORE-* confined	Concrete02	f'cc = 1.3 f'c; peak at 0.005; 50 % f'cc residual at 0.015	Wall boundary zones (0.6 m flanges, 1.0 m webs), code-compliant design
C-COL-5000 / 4000	Concrete02	Cover as unconfined; core f'cc = 1.2 f'c, peak at 0.004, 40 % residual at 0.012	Column fibre hinges
C-SLAB-4000	Concrete02 (beams); elastic (slab strips)	28 MPa	Model B beams; slab strips
SD40 bars	Steel02 (Menegotto-Pinto) inside MinMax	fy,exp = 1.1 x 400 = 440 MPa, E = 200 GPa, b = 0.01, R0 = 18, cR1 = 0.925, cR2 = 0.15; bar drops out at -1 % (unconfined, buckling) or -3 % (confined), and at +8 % (fracture)	All longitudinal bars

Expected vs specified strength. Nonlinear analysis uses expected strengths (steel 1.1 fy). Concrete uses the matrix level directly, because f'c is the deficiency variable: 3,000 psi stands for the low strength measured on site.

4. Reinforcement (new: not needed in ETABS)

Strength, yielding and failure all come from the bars, so every member needs reinforcement. The drawings are not public. The quantities are produced by a **design pass**: an elastic response-spectrum analysis to the DPT zone 5 spectrum (Fig. 12 values), with members sized to ACI 318-19 / DPT 1301/1302. The design pass is the same for Models A and B, and for every matrix level that changes the demand.

Member	Reinforcement	Source
Core walls, vertical	rho_v from flexure and axial demand, minimum 0.25 %	Design pass
Core walls, horizontal (shear)	rho_h for 2.7 x the RSA shear (ACI 18.10.3.1: 1.5 x 1.8), max 0.66 sqrt(f'c) on the section	Design pass, code-compliant
Wall boundary zones	Confined over 0.6-1.0 m (ties at about 100 mm)	Code-compliant
Link beams 250 x 800	Top and bottom bars from the RSA demand; 2-leg DB12 at 100 mm stirrups; required embedment ld per ACI 25.4	Design pass
Columns	2.5 % (F1-10), 2.0 % above	ETABS sheet Section 9
Beams 500 x 800 (Model B)	Top and bottom from the RSA plus gravity demand	Design pass
Slab (Model A)	Column-strip top steel; integrity bars through the column	Design pass, ACI 8.7.4.2

5. Elements (replaces ETABS shells and frames)

Component	OpenSees element	Key inputs	Failure modelled
Core wall piers (between openings)	forceBeamColumn per storey, fibre section of the whole pier (webs and flanges), aggregated with a shear spring; rigid arms to the floor nodes	Fibres per Section 3; shear spring HystereticSM: crack at 0.6 Vn, Vn (ACI Eq. 18.10.4.1) on 0.1 G Acv, held to 0.5 % shear drift, 20 % residual 0.25 % later (ASCE 41-17 Table 10-20)	Flexural crushing; shear failure; axial failure of a shear-damaged pier at 1-2 % drift (ASCE 41 'e')
Solid core (level A)	One closed-section pier per storey	St Venant torsion of the closed tube	As above; no link beams
Link beams 250 x 800	Elastic element with end plastic hinges (rotational springs)	My from bars; effective EI 0.15 EIg (ATC 72, as ETABS); ASCE 41-17 Table 10-19: theta_p 0.02 (flexure) or 0.012 (shear-controlled), residual 50 % / 30 %	Yield, shear failure; **anchorage pull-out** when embedment is short (bar stress capped at fy x le/ld, pull-out at 0.004 rad, 10 % residual)
Columns	forceBeamColumn, HingeRadau (hinge 0.5 D at each end), elastic middle	Fibre core and cover; -iter 30 1e-10	Flexural hinging; axial crushing
Beams (Model B)	Elastic T-beam with end hinges	0.35 EIg middle; ASCE 41-17 Table 10-7: theta_p 0.025, 20 % residual	Hinging; full moment connection to columns and core walls
Slab (Model A)	Effective-beam-width strips on every column line (Hwang & Moehle 2000)	Width 2c1 + l1/3 interior, c1 + l1/6 edge; cracked factor 0.5; hinges at the column-strip moment capacity	Flexural yielding (rarely governs)
Slab-column connection (Model A)	Two zeroLength elements between the slab node and the column node	(1) Stiff in shear and moment, **removed when it punches**: ACI 318-19 18.14.5.1 drift limit with the current Vg/Vc, or Vg >= Vc (Vc per ACI 22.6.5.2). (2) Post-punching vertical spring at 40 % Vc (columns), 30 % (walls)	**Punching** and punching cascade
Slab (Model B)	Not a separate element: the T-beam flange plus the diaphragm		None (no punching)
Floors	Rigid diaphragm (rigidDiaphragm) with a master node per floor carrying mass and rotary inertia	Semi-rigid in ETABS: compare mode shapes	-
P-Delta	PDelta geometric transformation on columns and wall piers	Not optional at 137 m	-

What happens to the stiffness modifiers. Fibre members crack and yield on their own, so 0.35 / 0.70 factors are not applied to them. Factors stay only where a member's middle part is modelled elastically: link beams 0.15, beams 0.35, slab strips 0.5. Check the cracked periods against ETABS; they should agree within about 10 %.

6. Supports, Connections and Deficiencies

Item	Setting	Note
Base	Fixed, all 6 DOF, at column bases and the wall pier bases	As ETABS 5.1; no soil springs (soft clay is in the input motion)
Column to slab (A)	Moment connection through the punching element	It transfers moment until it punches, then becomes a hinge with residual shear
Beam to column (B)	Fully fixed, no releases	As ETABS
Slab or beam to core wall	Continuous / fixed	As ETABS
f'c deficiency	Matrix levels A / B / C	3,000 psi = about 0.60 of the specified 5,000
Link-beam embedment (4th official finding)	Optional switch: le/ld = 1.0 (compliant) or 0.5	**Only possible in Phase 2**: bar stress capped at fy x le/ld, then pull-out
Non-compliant design (3rd official finding)	Optional switch: wall shear steel for the unamplified demand and unconfined boundaries	Alternative reinforcement set from the design pass

7. Seismic Demand: time histories instead of a response spectrum

The response spectrum is not applied to the model. The model is shaken by acceleration records, two horizontal components at once (X and Y), integrated step by step. The soft clay is still represented only through the input motion, which is a free-field surface record from the basin.

Input	Value	Source / note
Preferred records	Actual 28 March 2025 records: PWSA (deep basin, conservative) and PTNA (nearest to Chatuchak). Two horizontal components each	Available from the corresponding author (Ornthammarath, Mahidol University) on request, as stated in the paper
Fallback records	3 synthetic 2-component records, spectrally matched (5 %) to the Fig. 12 PWSA spectrum from 0.2 to 10 s; repeat for PTNA	Digitised spectra: etabs_response_spectra/ (same values as ETABS Function 2)
Duration	Significant duration D5-95 of about 133-156 s	Paper Table 2. Long-duration, long-period shaking must be preserved: it drives damage accumulation
Intensity SF = 1	The 2025 event as recorded (PGA about 0.02 g)	No scaling, no R
Code level	Records scaled so their 5 % spectrum matches DPT zone 5 DBE, then MCE, at T1 (0.2T1-1.5T1 average)	Replaces RS-DPT-U1/U2. DBE / MCE from Fig. 12 (corrected Function 1)
Collapse search (IDA)	SF ladder 0.5, 1, 1.5, 2, 3, 4.5, 6, 8, then bisection to 20 %	Collapse intensity = SF at collapse
Directions	Both components together; repeat with components swapped for selected runs	Replaces the SRSS directional combination
Accidental torsion	Not added	The core eccentricity is modelled explicitly; ETABS uses 0.05

Where the response spectrum still appears. (1) As the target the records are matched or scaled to. (2) In the design pass that sizes the reinforcement (Section 4). (3) In the modal and elastic check against ETABS. It is no longer how the earthquake load is applied.

8. Gravity, Mass and Damping

Item	Setting	Why
Gravity load	Self-weight (DEAD), plus SDL and 0.25 LL if adopted, applied in 10 load-control steps, then held constant (loadConst) before the earthquake	Wall axial load and slab shear at each column govern crushing and punching. Use the same gravity as the ETABS mass source so the two phases match
Mass	From the same loads; lumped at the floor master node (X, Y, rotation)	Matches ETABS MASS-SELFWEIGHT
Damping	**2.5 %** Rayleigh, anchored at T1 and max(T1/8, 0.35 s); not applied to the rigid zeroLength springs	Tall-building damping (DPT for H > 60 m; PEER TBI). Yielding adds hysteretic energy loss on top
Damping, sensitivity	5 % for selected runs	Shows how much the result depends on it

9. Analyses

Analysis	Settings	Purpose
Modal	Eigen, 12+ modes, after gravity	Periods and mass participation; check against ETABS
Gravity	Load control, 10 steps	Starting state for every other analysis
Pushover	Displacement control at the roof master node to 2.5 % drift; first-mode-like load pattern (h^2), X and Y, both signs	Strength, stiffness, sequence of yielding, ductility
Nonlinear time history	Newmark average acceleration; dt = 0.02 s (repeat at 0.01 s if a step fails numerically); KrylovNewton, fallbacks to Newton, NewtonLineSearch and ModifiedNewton, then sub-steps; NormDispIncr 1e-5	Response to the 2025 event and to the code levels
IDA	Section 7 ladder	Collapse intensity
Sudden core-pier removal	Remove pier elements under gravity in one step; 5 % damping; 6 s	Alternate load path: does the floor system hold?
Pushdown	Withdraw the removed pier's force gradually	Static capacity of the alternate path, and the force carried by each path

10. Collapse Criteria and Outputs

Item	Definition
Collapse	Any one of: storey drift at a corner > 10 %; floor vertical drop > 0.6 m; or loss of convergence after a wall, column or punching failure. Numerical stops are repeated at dt = 0.01 s; a second stop counts as collapse
Damage events (time-stamped)	Link-beam yield and failure, wall shear strength loss, wall axial failure, column failure, punching, with location
Outputs per run	Peak and residual drift per storey; base shear (core vs columns); torsion; pier demand/capacity; gravity-load path before and after damage; event sequence

11. To Resolve Before Building

Item	Issue	Suggested resolution
Reinforcement	Not public; nonlinear results depend on it	Design pass to the Fig. 12 DPT spectrum, same rules for A and B; state it in the methodology
Records	Actual 2025 records not yet obtained	Request them. Until then use synthetic records matched to the Fig. 12 PWSA spectrum, with the recorded duration
Slab type, Model A	The outline gives a 4,000 psi slab; the investigation describes a post-tensioned flat slab	Decide PT or RC: it changes punching strength (PT: 0.29 sqrt(f'c) + 0.3 fpc) and the drift limit
Link-beam depth	800 mm here. The current Phase 2 runs use 1.7 m (door spandrel), which gave periods of 3.0-3.2 s	Use 800 mm to match ETABS, and keep 1.7 m as a sensitivity
Grid	7 lines at 6.5 m here. The current Phase 2 runs use 6 lines	Rebuild the Phase 2 model on the 7-line grid so both phases share it
Diaphragm	Semi-rigid in ETABS, rigid in OpenSees	Acceptable: compare modes; the vertical redistribution after pier loss is carried by the slab strips and beams, not the diaphragm
Run count	54 models x records x intensities	Stage the runs as in Section 2